

=== 8-Puzzle Solver ===

Enter numbers 0 to 8 separated by space (0 represents the blank space).

Enter initial state (e.g., 1 2 3 4 0 6 7 5 8): 1 2 3 0 5 6 4 7 8

Enter goal state (press Enter for default '1 2 3 4 5 6 7 8 0'): 1 2 3 4 5 6 7 8 0

Initial State:

1 2 3

_ 5 6

4 7 8

Goal State:

1 2 3

4 5 6

7 8 _

Choose Search Algorithm:

1. Depth-First Search (DFS)

2. Iterative Deepening Search (IDS)

Enter option (1 or 2): 1

Running DFS...

Solution found in 19 steps!

Move sequence: Right -> Right -> Up -> Left -> Down -> Right -> Up -> Left -> Down
-> Left -> Down -> Right -> Up -> Right -> Up -> Left -> Down -> Down -> Right

Step-by-step path:

Step 0:

1 2 3

_ 5 6

4 7 8

Step 1:

1 2 3

5 _ 6

4 7 8

Step 2:

1 2 3

5 6 _

4 7 8

Step 3:

1 2 _

5 6 3

4 7 8

Step 4:

1 _ 2

5 6 3

4 7 8

Step 5:

1 6 2

5 _ 3

4 7 8

Step 6:

1 6 2

5 3 _

4 7 8

Step 7:

1 6 _

5 3 2

4 7 8

Step 8:

1 _ 6

5 3 2

4 7 8

Step 9:

1 3 6

5 _ 2

4 7 8

Step 10:

1 3 6

_ 5 2

4 7 8

Step 11:

1 3 6

4 5 2

_ 7 8

Step 12:

1 3 6

4 5 2

7 _ 8

Step 13:

1 3 6

4 _ 2

7 5 8

Step 14:

1 3 6

4 2 _

7 5 8

Step 15:

1 3 _

4 2 6

7 5 8

Step 16:

1 _ 3

4 2 6

7 5 8

Step 17:

1 2 3

4 _ 6

7 5 8

Step 18:

1 2 3

4 5 6

7 _ 8

Step 19:

1 2 3

4 5 6

7 8 _

=== 8-Puzzle Solver ===

Enter numbers 0 to 8 separated by space (0 represents the blank space).

Enter initial state (e.g., 1 2 3 4 0 6 7 5 8): 1 2 3 0 5 6 4 7 8

Enter goal state (press Enter for default '1 2 3 4 5 6 7 8 0'): 1 2 3 4 5 6 7 8 0

Initial State:

1 2 3

_ 5 6

4 7 8

Goal State:

1 2 3

4 5 6

7 8 _

Choose Search Algorithm:

1. Depth-First Search (DFS)

2. Iterative Deepening Search (IDS)

Enter option (1 or 2): 2

Running IDS...

Optimal solution found at depth 3 with 3 steps!

Move sequence: Down -> Right -> Right

Step-by-step path:

Step 0:

1 2 3

_ 5 6

4 7 8

Step 1:

1 2 3

4 5 6

_ 7 8

Step 2:

1 2 3

4 5 6

7 _ 8

Step 3:

1 2 3

4 5 6

7 8 _